

**AMENDMENT NO. 1 DECEMBER 1992
TO
IS 12915 : 1990 ACEPHATE, TECHNICAL —
SPECIFICATION**

(*Page 2, clause A-1.1*) — Substitute the following for the existing clause:

'A-1.1 Two methods have been specified for the determination of acephate content, namely, gas chromatographic (GLC) method (*see A-2*) and phosphorus method (*see A-3*). The GLC method shall be followed by independent laboratories. The phosphorus method shall be followed by the manufacturer only for routine in-plant quality control checks.'

(*Page 2*) — Insert the following at the end:

'A-3 PHOSPHORUS METHOD

A-3.1 Apparatus

A-3.1.1 Shaking machine

A-3.2 Reagents

A-3.2.1 Ammonium Nitrate Solution — Dissolve 100 g ammonium nitrate (free of phosphates) in water and dilute to 1 000 ml.

A-3.2.2 Methyl Orange Indicator Solution — 0.1 percent aqueous solution (v/v).

A-3.2.3 Standard Sodium Hydroxide Solution — 0.1 N.

A-3.2.4 Standard Hydrochloric Acid Solution — 0.1 N.

A-3.2.5 Phenolphthalein Indicator — 1 percent solution in ethyl alcohol.

A-3.2.6 Ammonium Hydroxide Solution — Aqueous solution (1:3; v/v).

A-3.2.7 Nitric Acid Solution — Aqueous solution (1:3;v/v).

A-3.2.8 Ammonium Molybdate Solution — Dissolve 100 g ammonium molybdate in a mixture of 144 ml ammonium hydroxide and 271 ml water. Cool to room temperature and add slowly with constant stirring to a cool mixture of 489 ml nitric acid and 1148 ml water. Allow the mixture to stand for 48 hours in a stoppered vessel. Filter immediately before use and add 5 ml nitric acid to each 100 ml of filtrate.

A-3.2.9 Chloroform

A-3.2.10 Perchloric Acid — 70-72 percent (v/v).

A-3.3 Procedure

A-3.3.1 Accurately weigh the sample containing 200 mg phosphorus and extract with 200 ml chloroform by shaking for one hour on a shaking machine. Filter into a 250-ml volumetric flask, wash the residue twice with 20 ml portions of chloroform and make up the volume with chloroform.

A-3.3.2 Transfer the aliquot into a 250-ml beaker, add 30 ml nitric acid, dropwise at first until vigorous reaction is completed, boil gently for 30 minutes. Cool and add perchloric acid.

A-3.3.3 Boil gently the solution at A-3.3.2 until it is colourless or nearly so and appearance of dense fumes. Do not boil the solution to dryness. The fuming temperature shall be at about 170°C for at least 30 minutes. Add more quantity of perchloric acid, if necessary. Cool slightly, add 50 ml water and boil for a few minutes.

A-3.3.4 Add to the solution (A-3.3.3), 20 ml ammonium nitrate solution and 1.0 ml methyl orange indicator. Adjust the pH to just alkaline by adding ammonium hydroxide and then to acidic with addition of nitric acid. Dilute to 100 ± 20 ml with distilled water.

A-3.3.5 Add ammonium molybdate solution (A-3.2.8) to the solution (A-3.3.4) and agitate continuously for 15 minutes. Allow the precipitate to settle for one or two minutes.

A-3.3.6 Decant the supernatant liquid as at A-3.3.5 through a glass funnel fitted with a pad of macerated asbestos approximately 0.31 cm thick on a perforated disc and filter under vacuum. Transfer the precipitate to the funnel and rinse the beaker several times with a stream of water. Wash the precipitate 5 times with 25-30 ml water. Transfer the precipitate, pad and perforated disc back into the 250-ml beaker and rinse the funnel with not more than 100 ml water.

A-3.3.7 Add sufficient standard sodium hydroxide to completely discharge the yellow phospho-molybdate and then add 2-3 ml in excess. Add 0.4 ml phenolphthalein indicator. Rinse the sides of the beaker with 5-10 ml sodium hydroxide. Titrate with standard hydrochloric acid to a distinct light pink colour and point and verify the end point at 7.85 ± 0.05 pH by means of a pH meter.

A-3.4 Calculation

$$\text{Phosphorus content, percent by mass } P = \frac{(V_1 \times N_1) - (V_2 \times N_2)}{M} \times 0.1349$$

where

V_1 = volume, in ml, of standard sodium hydroxide;

N_1 = normality of standard sodium hydroxide;

V_2 = volume, in ml, of standard hydrochloric acid

N_2 = normality of standard hydrochloric acid; and

M = mass, in g, of sample in aliquot (A-3.2.2).

Acephate content, percent by mass = $P \times 5.91$ '.

(FAD 1)

Reprography Unit, BIS, New Delhi, India